🎮 “Mix Match Memory Quest”

This is a **3-card matching memory game** where players must form a **complete set**:

> ✅ **1 Term + 1 Picture + 1 Description = 1 Correct Match Set**

Example of a correct set:

- **Uniform Mixture** (term)
- **Salt Water** (picture)
- **“Looks the same throughout”** (description)

🧠 GAME LOOP (Step-by-Step Gameplay Flow)

1. Start Phase

- Player enters the **pixel lab room**
- Grid of **face-down cards** appears (e.g., 12–18 cards)
- Timer starts (e.g., 90 seconds)

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2. Player Action Loop

Player can only flip **3 cards at a time**

Step 1: Flip Cards

- Player taps a card → it flips (animation)
- Player flips **up to 3 cards**

Step 2: System Check (Core Logic)

After 3 cards are flipped:

👉 The system checks:

IF (1 Term + 1 Picture + 1 Description AND all belong to same type)
→ CORRECT MATCH
ELSE
→ WRONG MATCH

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3. Feedback Phase

✅ **If Correct:**

- Cards glow + sparkle ✨
- Cards stay revealed (locked)
- Score increases (+10 or +20)
- “Science Clue” pops up:

  “Uniform mixtures look the same throughout.”

- Lab background lights up (progress feedback)

❌ **If Wrong:**

- Cards shake ❌
- Flip back after 1–2 seconds
- Small hint appears (optional):

  “Check the appearance carefully!”

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4. Progression System

- Each correct set = **1 power bar filled**
- When bar fills:
  → **New set unlocks** OR level completes

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5. Win Condition

Player wins when:

- All sets are matched **before time runs out**

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6. Lose Condition

Player loses when:

- Timer reaches 0 and sets are incomplete

🧩 CARD TYPES (Game Data Structure)

You will have **3 categories of cards**:

1. Term Cards

- Uniform Mixture
- Non-uniform Mixture

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2. Picture Cards

- Orange Juice
- Salt Water
- Sand and Water
- Fruit Salad

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3. Description Cards

- “Looks the same throughout”
- “Components can be seen separately”

🔁 MATCHING LOGIC (Important for system design)

Each card should have a hidden **ID or tag**:

Example:

Set ID: 001 (Uniform)
- Term: Uniform Mixture
- Picture: Salt Water
- Description: Looks the same throughout

Game checks:

IF all flipped cards have SAME Set ID
→ Correct

🎯 SCORING SYSTEM

Action	Points
Correct match	+20
Fast match bonus	+5
Wrong match	0
Finish early	+50

⚡ CORE GAME MECHANICS SUMMARY

- Flip cards (memory mechanic)
- Match **3 related items**
- Time pressure
- Immediate feedback (visual + text)
- Progressive unlocking
- Reinforcement via clues

🎮 PLAYER EXPERIENCE FLOW

1. Sees grid of mystery cards
2. Flips cards → curiosity
3. Thinks: “Do these belong together?”
4. Gets feedback → learns concept
5. Repeats → reinforces memory
6. Wins → feels mastery

🧪 WHY THIS WORKS (For your defense)

This mechanic supports:

✔ Remembering

- recalling terms and definitions

✔ Understanding

- connecting:
    - appearance
    - example
    - concept

✔ Cognitive Reinforcement

- repetition through gameplay
- immediate feedback loop
- visual + textual learning